The Hertzian Dipole

James T. Aberle, Ph.D.
Associate Professor Emeritus

School of ECEE

Ira A. Fulton Schools of Engineering
Arizona State University

Lecture Objective

By the end of this lecture, you will be able to:

- Understand the characteristics of the Hertzian dipole and its usefulness as a building block of wire antennas,
- Evaluate the electromagnetic field radiated by the Hertzian dipole,
- Analyze the antenna properties of the Hertzian dipole.

What is an Hertzian Dipole?

- A **Hertzian dipole** (also called an *elemental dipole* or *infinitesimal dipole)* is an ideal theoretical concept for a point source of electromagnetic radiation.
- The Hertzian dipole is an infinitesimally short straight perfect conductor carrying a time-harmonic (sinusoidal) current that does not vary with position on the dipole.

The Hertzian Dipole

- The current on the dipole is given by

$$i(t) = I_0 \cos \omega t, \ -l/2 \leq z \leq l/2$$

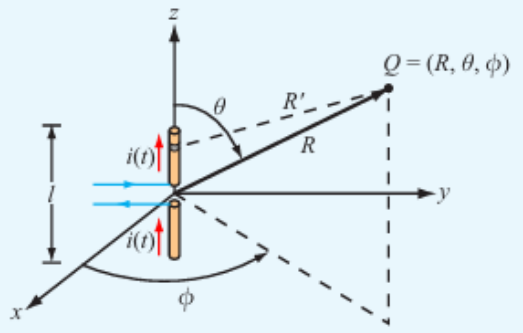


Figure 9-4 Short dipole placed at the origin of a spherical coordinate system.

Question

The assumed current distribution on the Hertzian dipole is not actually physically realizable. Why?

a) The current does not go to zero at the open end.
b) The current varies sinusoidally as a function of time.
c) The wire is shorter than one wavelength.
d) Current flows in closed loops.

Phasor Retarded Vector Potential (1 of 2)

- Recall that the phasor vector magnetic potential for a time-harmonic current source is given by

$$\tilde{\vec{A}}(\vec{R}) = \frac{\mu_0}{4\pi} \int_{V'} \tilde{\vec{J}}(\vec{R}') \frac{e^{-k_0 R'}}{R'} dv'$$

- Applying the above equation to the Hertzian dipole, we have

$$\tilde{\vec{A}}(\vec{R}) = \frac{\mu_0}{4\pi} \int_{-l/2}^{l/2} \hat{z} I_0 \frac{e^{-k_0 R'}}{R'} dz'$$

Phasor Retarded Vector Potential (2 of 2)

- Let's assume that we are far enough away from the very short dipole so that

$$R' \approx R$$

- We have

Spherical propagation (wave) factor

$$\tilde{\vec{A}}(\vec{R}) = \hat{z}\frac{\mu_0}{4\pi} I_0 l \left(\frac{e^{-k_0 R}}{R}\right)$$

Derivation of Electric and Magnetic Fields of Hertzian Dipole (1 of 2)

1. Obtain spherical coordinate representation of vector potential using

$$\hat{z} = \hat{R}\cos\theta - \hat{\theta}\sin\theta$$

$$\tilde{A}_R = \frac{\mu_0}{4\pi} I_0 l \cos\theta \left(\frac{e^{-k_0 R}}{R}\right)$$

$$\tilde{A}_\theta = -\frac{\mu_0}{4\pi} I_0 l \sin\theta \left(\frac{e^{-k_0 R}}{R}\right)$$

$$\tilde{A}_\phi = 0$$

Derivation of Electric and Magnetic Fields of Hertzian Dipole (2 of 2)

2. Use the equations relating electric and magnetic fields to vector magnetic potential

$$\tilde{\vec{H}} = \frac{1}{\mu_0} \nabla \times \tilde{\vec{A}}$$

$$\tilde{\vec{E}} = \frac{1}{j\omega\varepsilon_0} \nabla \times \tilde{\vec{H}}$$

Derivation of Electric and Magnetic Fields of Hertzian Dipole

$$\widetilde{H}_{\phi} = I_0 l \frac{{k_0}^2}{4\pi} \sin\theta \, e^{-k_0 R} \left(\frac{j}{k_0 R} + \frac{1}{(k_0 R)^2} \right)$$

$$\tilde{E}_R = I_0 l \frac{{k_0}^2 \eta_0}{2\pi} \cos\theta \, e^{-k_0 R} \left(\frac{1}{(k_0 R)^2} - \frac{j}{(k_0 R)^3} \right)$$

$$\tilde{E}_{\theta} = I_0 l \frac{{k_0}^2 \eta_0}{4\pi} \sin\theta \, e^{-k_0 R} \left(\frac{j}{k_0 R} + \frac{1}{(k_0 R)^2} - \frac{j}{(k_0 R)^3} \right)$$

Other components of $\vec{E}$ and $\vec{H}$ are zero.

Far-Field Approximation

$$k_0 R = 2\pi \frac{R}{\lambda} \gg 1$$

$$\tilde{E}_\theta = I_0 l \frac{j k_0 \eta_0}{4\pi} \sin\theta \left(\frac{e^{-k_0 R}}{R} \right)$$

$$\tilde{E}_R \approx 0$$

$$\tilde{H}_\phi = I_0 l \frac{j k_0}{4\pi} \sin\theta \left(\frac{e^{-k_0 R}}{R} \right) \quad = \frac{\tilde{E}_\theta}{\eta_0}$$

Electric Field Surrounding Hertzian Dipole

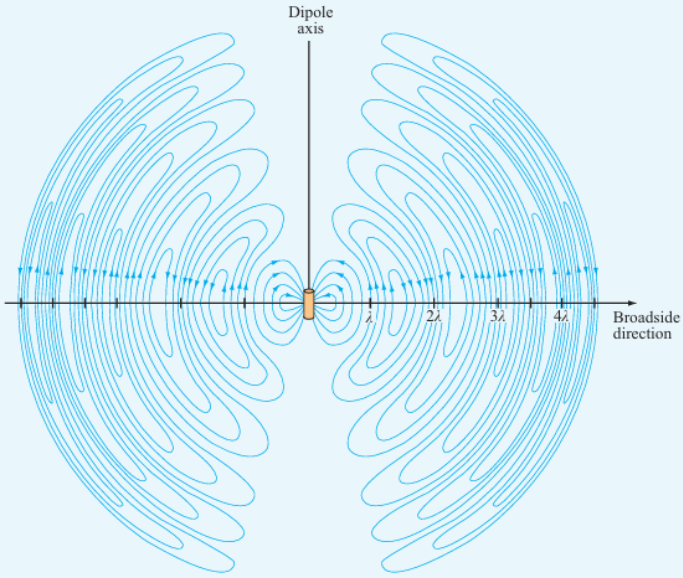


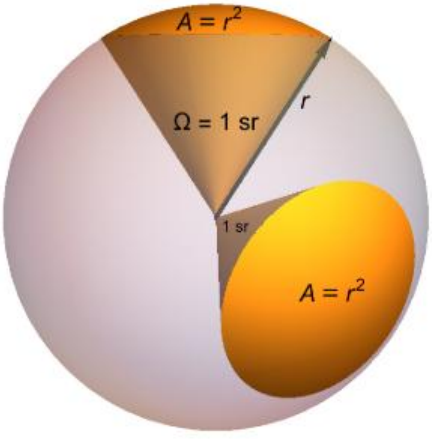
Figure 9-6 Electric field lines surrounding an oscillating dipole at a given instant.

Radian and Steradian

- One *radian* is the plane angle on a circle subtended by an arc whose length is equal to the radius of the circle. There are 2π radians in a circle.
- One *steradian* is the solid angle in a sphere subtended by a spherical surface equal to the radius squared of the sphere. There are 4π steradians in a sphere.

Visualization of Steradian

- Imagine a sphere.
- Draw a cone from the center of the sphere and let it cut out an area on the sphere.
- The **solid angle** of the cone is a measure of how “wide” the cone is.
- If the area cut out by the cone has an area equal to r^2 ,then the cone’s solid angle is 1 steradian (sr).



https://upload.wikimedia.org/wikipedia/commons/5/5b/Solid_Angle%2C_1_Steradian.svg

Infinitesimal Elements of Surface Area and Solid Angle

- The infinitesimal area element on the surface of a sphere is given by

$$dA = R^2 \sin\theta \, d\theta \, d\phi = R^2 d\Omega$$

- The infinitesimal element of solid angle is given by

$$d\Omega = \frac{dA}{R^2} = \sin\theta \, d\theta \, d\phi$$

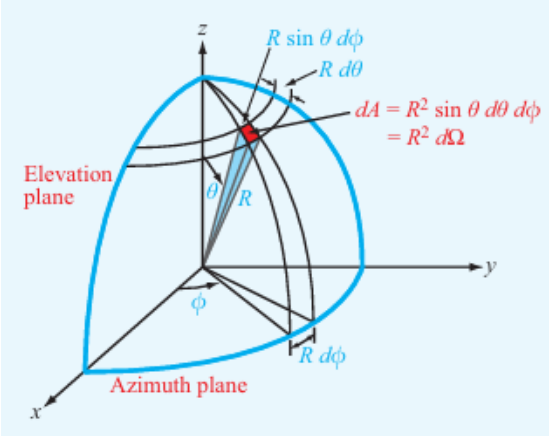


Figure 9-8 Definition of solid angle $d\Omega = \sin\theta \, d\theta \, d\phi$.

Time-Average Poynting Vector

- Time-Average ***Poynting vector*** has units of *watts per square meter* and represents time-average magnitude and direction of power density flow:

$$\vec{S}_{\mathrm{av}}(\vec{R}) = \frac{1}{2}\mathrm{Re}\left(\tilde{\vec{E}} \times \tilde{\vec{H}}^{*}\right)$$

W/m^2 V/m A/m

Time-Average Poynting Vector of Hertzian Dipole

$$\vec{S}_{\mathrm{av}}(\vec{R}) = \frac{1}{2}\mathrm{Re}\left(\tilde{\vec{E}} \times \tilde{\vec{H}}^*\right) \quad = \hat{R}\,\frac{\left|\tilde{E}_\theta\right|^2}{2\eta_0}$$

$$= \hat{R}\left(\frac{\eta_0 {k_0}^2}{32\pi^2}\right)(I_0 l)^2\,\frac{1}{R^2}\sin^2\theta$$

Radiation Intensity

- ***Radiation intensity*** is "the power radiated from an antenna per unit solid angle."
- The *radiation intensity* is a far-field quantity and has units of *watts per steradian*.
- The radiation intensity is a function of θ and ϕ (i.e., of direction) but not of R.

Radiation Intensity of Hertzian Dipole

- The radiation intensity is defined by

$$U(\theta,\varphi) = \lim_{R\to\infty} R^2\, S_{\mathrm{av}}(R,\theta,\varphi)$$

$$\propto \frac{1}{R^2}$$

$$= \left(\frac{\eta_0 {k_0}^2}{32\pi^2}\right)(I_0 l)^2 \sin^2\theta$$

Normalized Radiation Intensity

- The ***normalized radiation intensity*** is the ratio of the radiation intensity in a given direction to the maximum radiation intensity

$$\boxed{F(\theta,\varphi) = \frac{U(\theta,\varphi)}{U_{\max}}}$$

Normalized Radiation Intensity of Hertzian Dipole

$$F(\theta,\varphi) = \sin^2\theta$$

The Hertzian dipole exhibits an omnidirectional radiation pattern

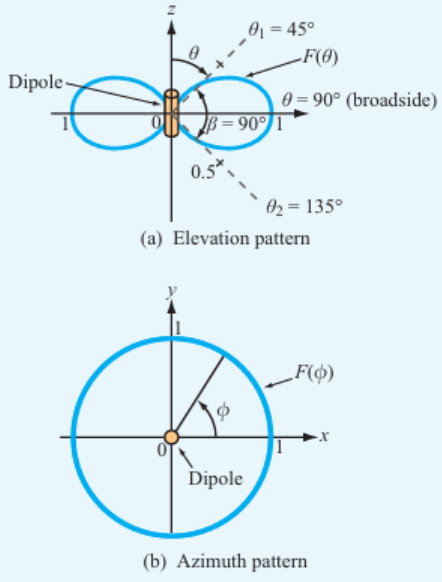


Figure 9-7 Radiation patterns of a short dipole.

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